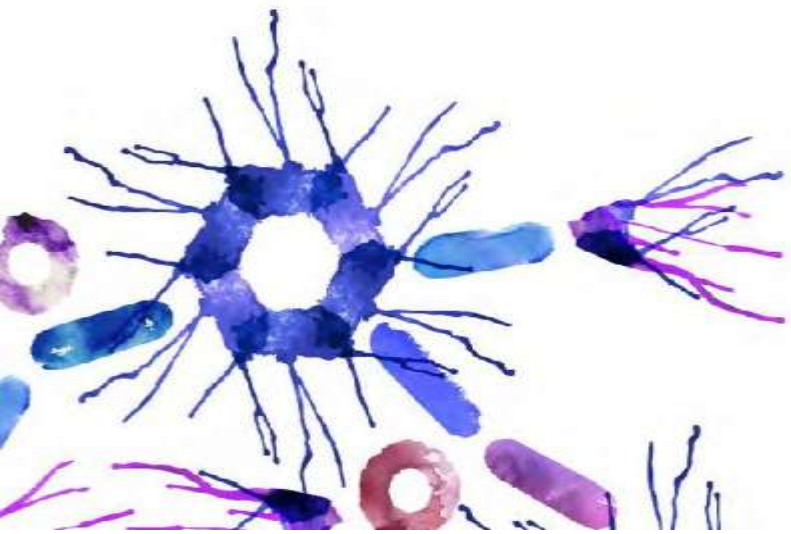


Epidemiology and Prognosis of MS

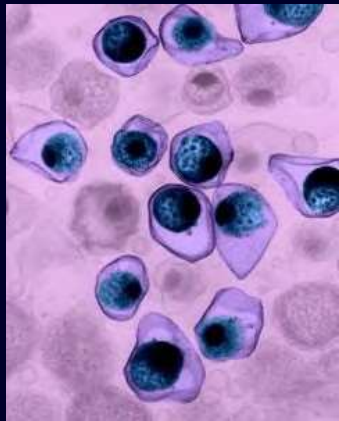
Dr Ben Turner



Potential Triggers for Multiple Sclerosis



Infectious agent



Genetic
predisposition



Environmental
factors

Abnormal immunologic response ➔ **MS**

Epidemiology

- Age of onset - 3rd / 4th decade (16 - 50 years)
- Prevalence – UK ~125/100,000 (latitude dependent)
- In an unpublished communication, Prof. Bohlega estimated the prevalence of MS in Saudis to be 40/100,000 in 2008.
- Life Span - slightly reduced (~ 10 years)
- Sex - F > M
- Race - Caucasians
(uncommon in Chinese / ? Viking ancestral genes)
- Geography - Northern European Disease

PREVALENCE OF MULTIPLE SCLEROSIS

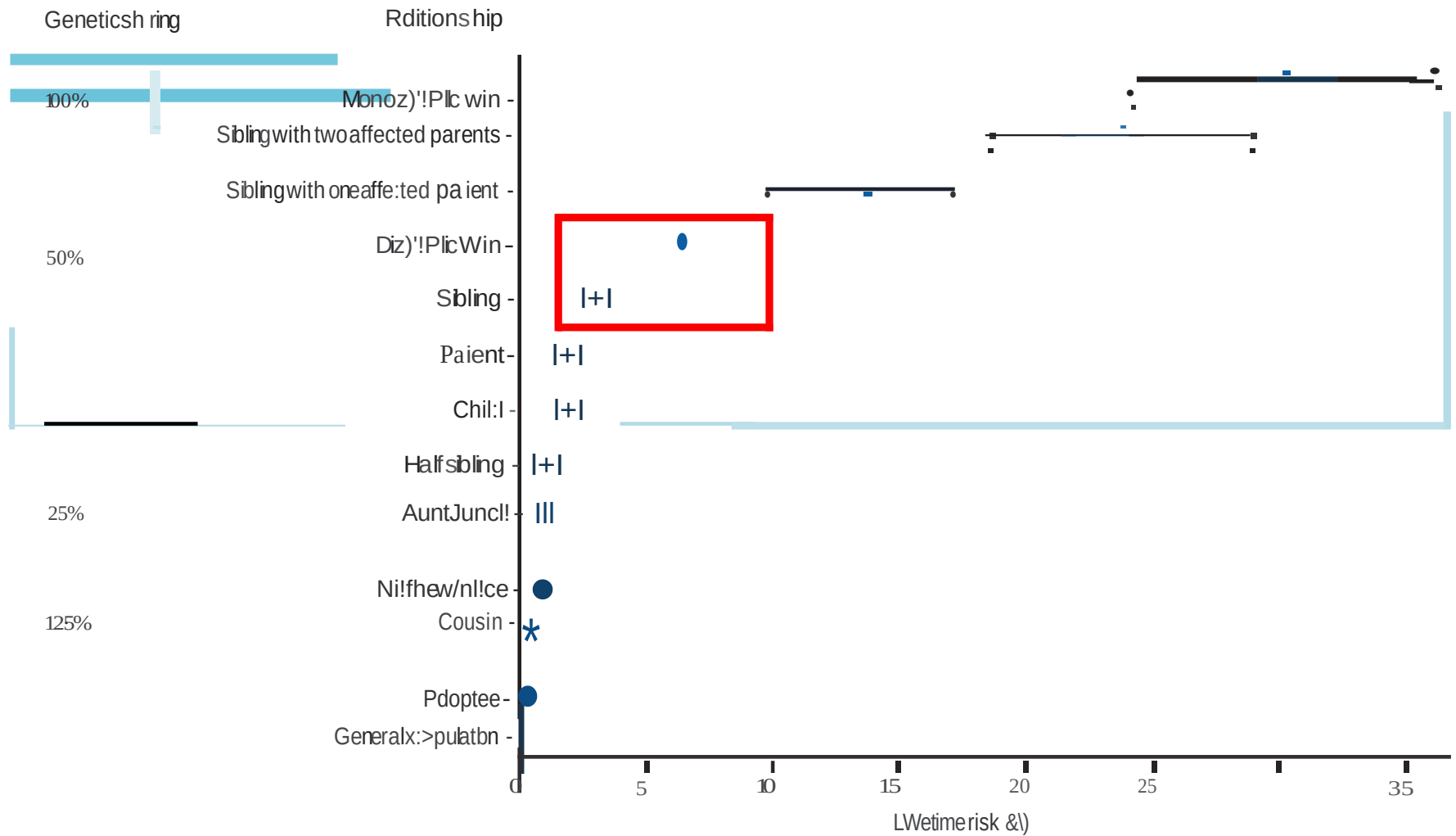
Prevalence of MS per 100,000 population



Note: 93 countries replied to questions providing data for this map

SOURCE: MSIF/WHO

Familial Risk



The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812

AUGUST 30, 2007

VOL. 357 NO. 9

Risk Alleles for Multiple Sclerosis Identified by a Genomewide Study

The International Multiple Sclerosis Genetics Consortium*

HLA-DRB1

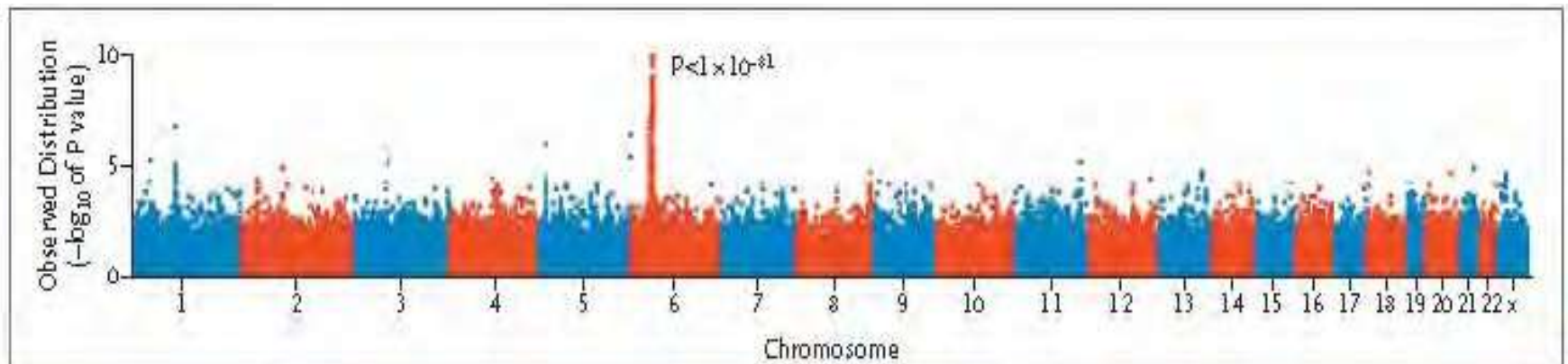


Figure 2. Overview of the Primary Genomewide Association Scan Involving 931 Family Trios.

P values (shown as $-\log_{10}$ values) for results of transmission disequilibrium testing are plotted across the genome. The classic HLA-DR risk locus on chromosome 6p21 stands out with strong statistical significance ($P < 1 \times 10^{-81}$).

Other Genetic contributions

HLA-DRB1 15.01 (polymorphism) OR = 3.1

The MS Genome 2012

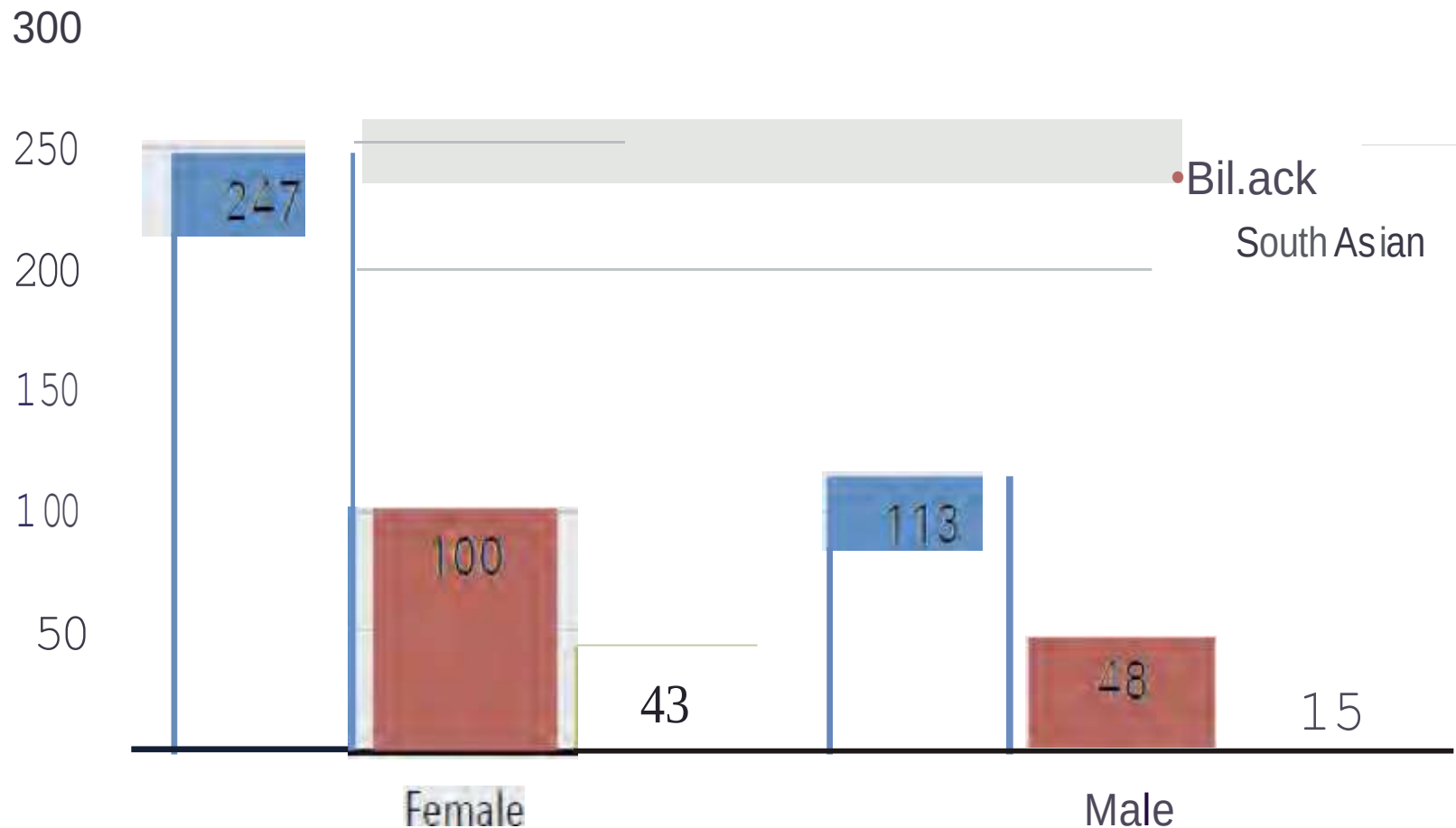
-> 100 non HLA-DRB1 polymorphism OR = 1.1 to 1.2

Odds ratio's small, few %

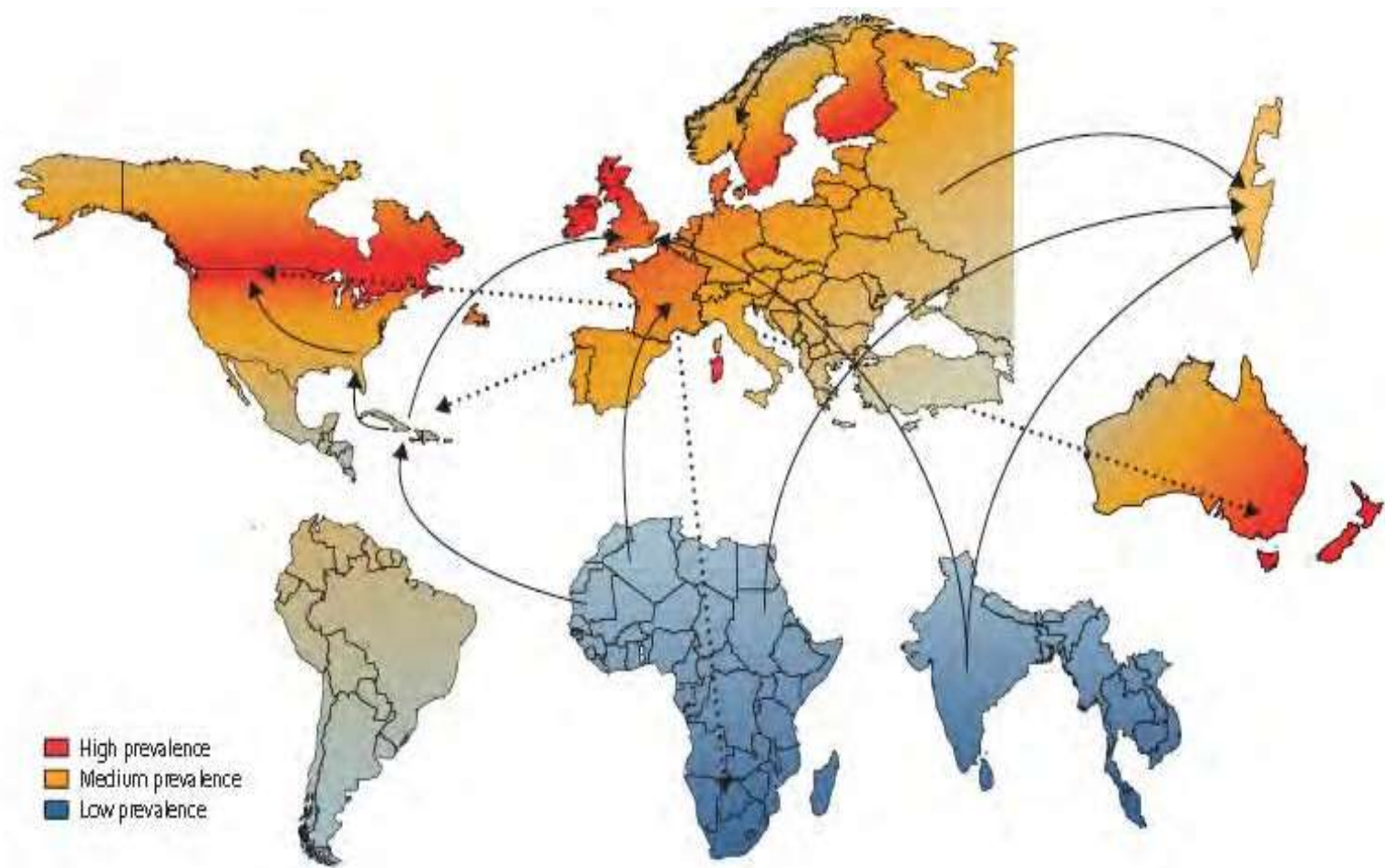
Genes (15to20% influence risk)

Barts MS ethic distribution

Sex and Ethnic-specific MS Prevalence (per 100K)



Migration studies



Increasing female incidence

The changing demographic pattern of multiple sclerosis epidemiology

Nikolaj Koch-Henriksen, Per Søberg Shøer

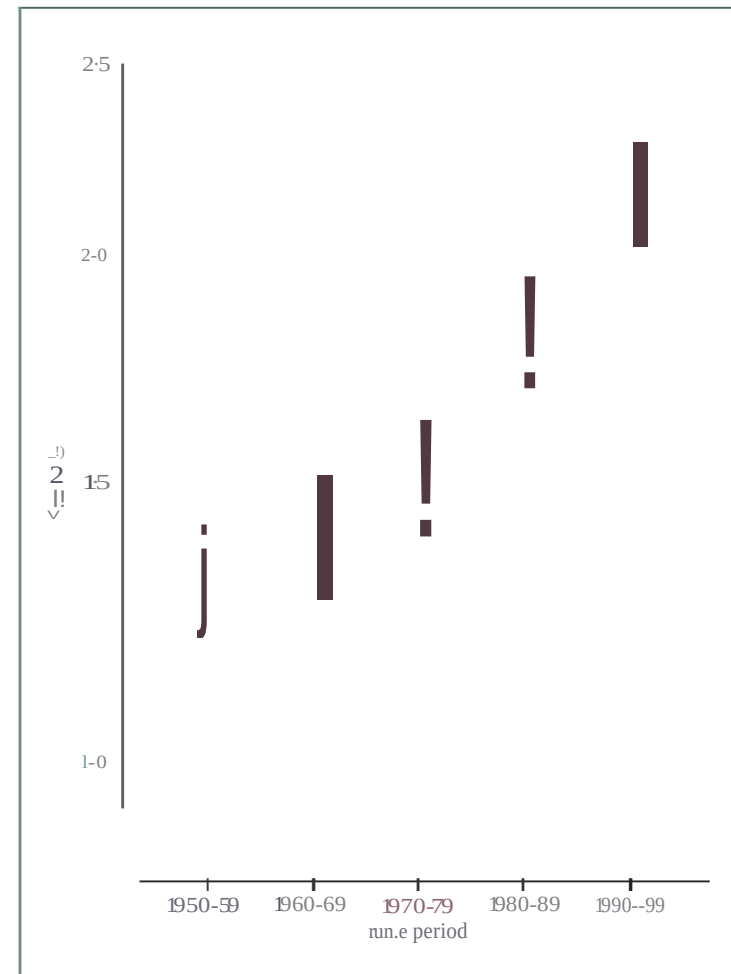
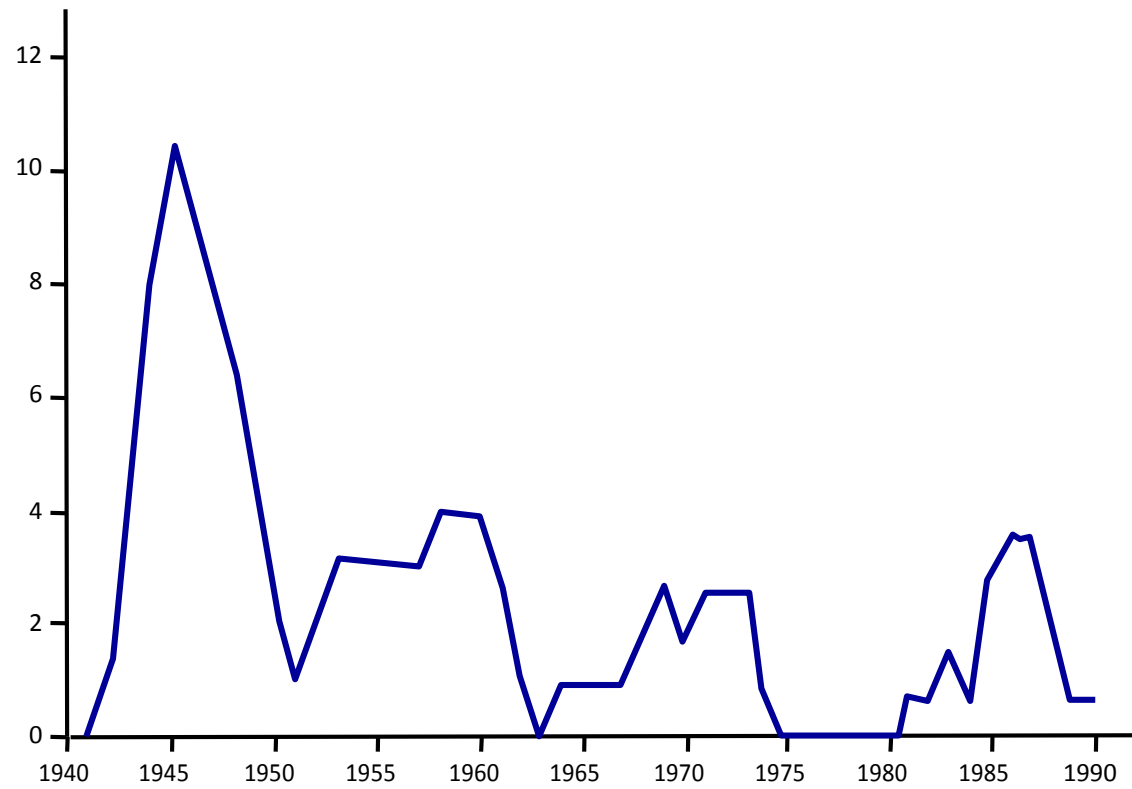


Figure 8: Increase in female: male sex ratio in Denmark since 1950
 A marked increase in incidence of multiple sclerosis in women can be seen. Sex ratios (95% CIs) are shown on a logarithmic scale. Sex ratios are calculated as the ratio of life-time cumulative incidence proportions compiled on the basis of age-specific and sex-specific incidence rates in a given period. This method replaces standardisation and makes ratios at different times comparable.

Epidemics or clusters of MS

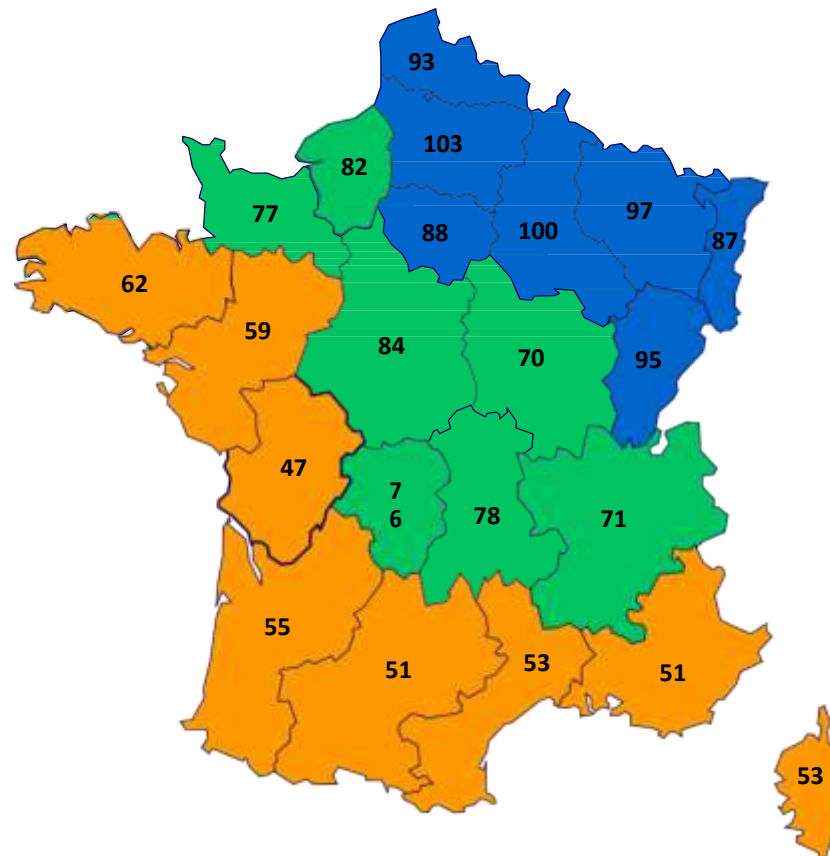
The annual incidence of MS (per 100 000 inhabitants) in the Faroe Islands since 1940



Enviromental Risk Factors

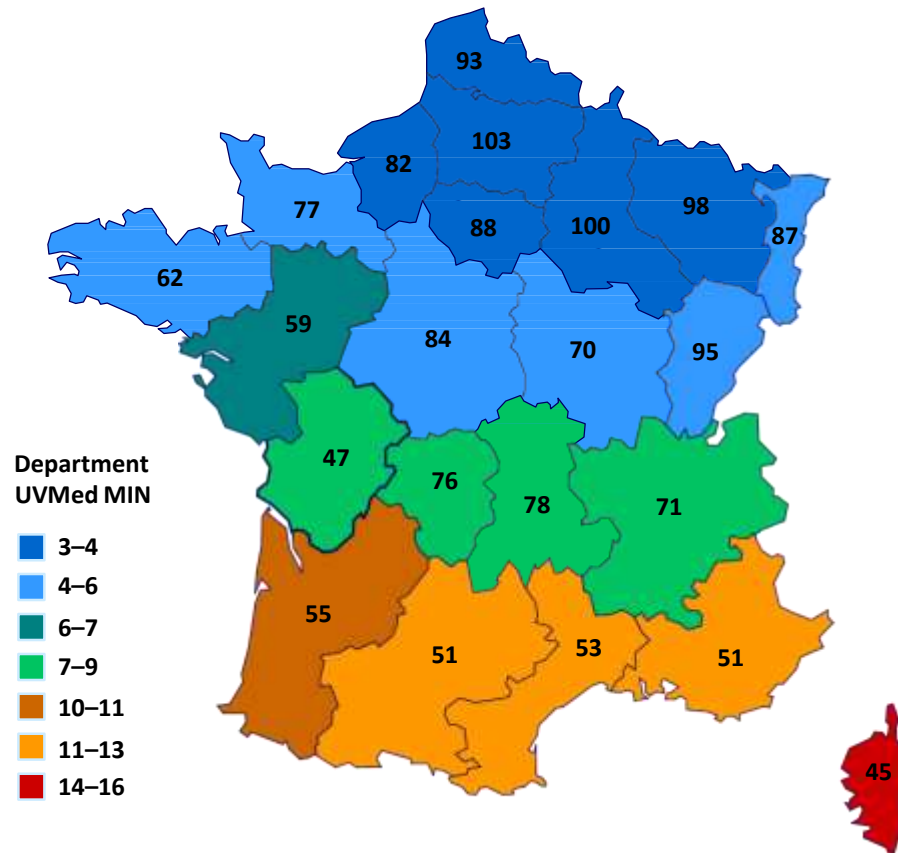
- Sunlight/UVB
- Vitamin D insufficiency (x2 risks)
- Remote EBV (x 2-3 risk) 10% of paediatric MS EBV neg
- Smoking (increases 50 to 70%)
- Parity (decreased risk of no. of children) cumulative beneficial effect of pregnancy
- Obesity (lower Vit D)
- Salt – pro-inflammatory, not confirmed

Geographical distribution of MS: prevalence increases away from the equator



Role of vD_3 : UVB and MS prevalence

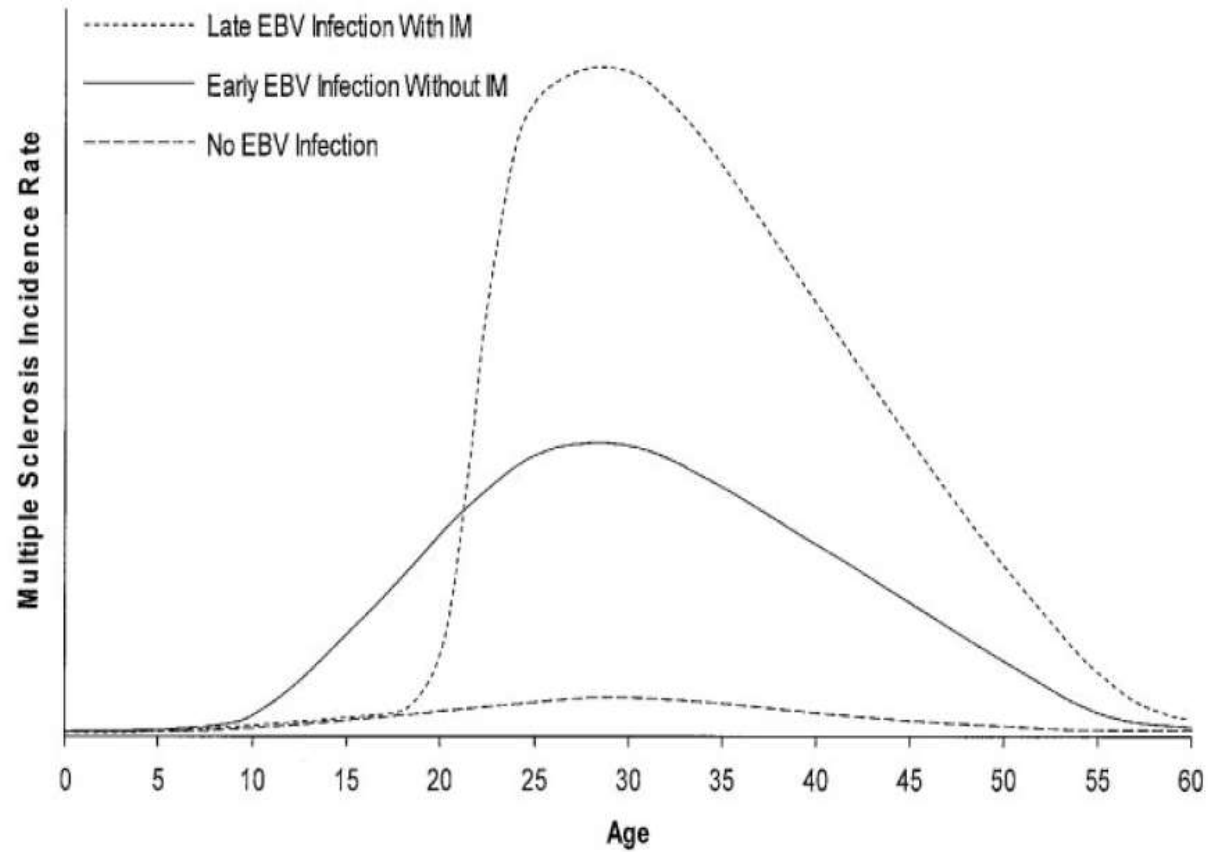
MS Prevalence by Department Against UVMED minimum



¹Jablonski NG, Chaplin G. *J Hum Evol* 2000;**39**:57–106.

²Chaplin G. *Am J Phys Anthropol* 2004;**125**:292–302.

Symptomatic Epstein-Barr virus (EBV) infection and MS



Common viruses associated with lower pediatric multiple sclerosis risk.

Table 5 Multivariate model including all available remote viral exposures (adjusted for age at blood draw, sex, race, and ethnicity)			
	OR to develop MS/CIS	95% CI	p Value
Anti-EBNA-1-positive	5.00	1.80-13.90	0.002
Anti-CMV-positive	0.30	0.11-0.77	0.01
Anti-HSV-1-positive	2.86	0.75-10.89	0.12
DRB1*1501/1503-positive	3.00	1.27-7.11	0.01

Abbreviations: CI = confidence interval; CIS = clinically isolated syndrome; CMV = cytomegalovirus; EBNA = Epstein-Barr nuclear antigen; HSV = herpes simplex virus; MS = multiple sclerosis; OR = odds ratio.

Conclusions: These findings suggest that some infections with common viruses may in fact lower MS susceptibility. If this is confirmed, the pathways for risk modification remain to be elucidated. E.Waubant et al *Neurology*® 2011;76:1989–1995

‘Hygiene Hypothesis’ of autoimmune disease

Ponsonby AL, van der Mei I, Dwyer T, et al. Exposure to infant siblings during early life and risk of multiple sclerosis. JAMA 2005;293:463–9

Pedrini MJF, Seewann A, Bennett KA, et al. Helicobacter pylori as a protective factor against multiple sclerosis risk in females. J Neurol Neurosurg Psychiatry 2015;86:603–7

Gene-environment interactions

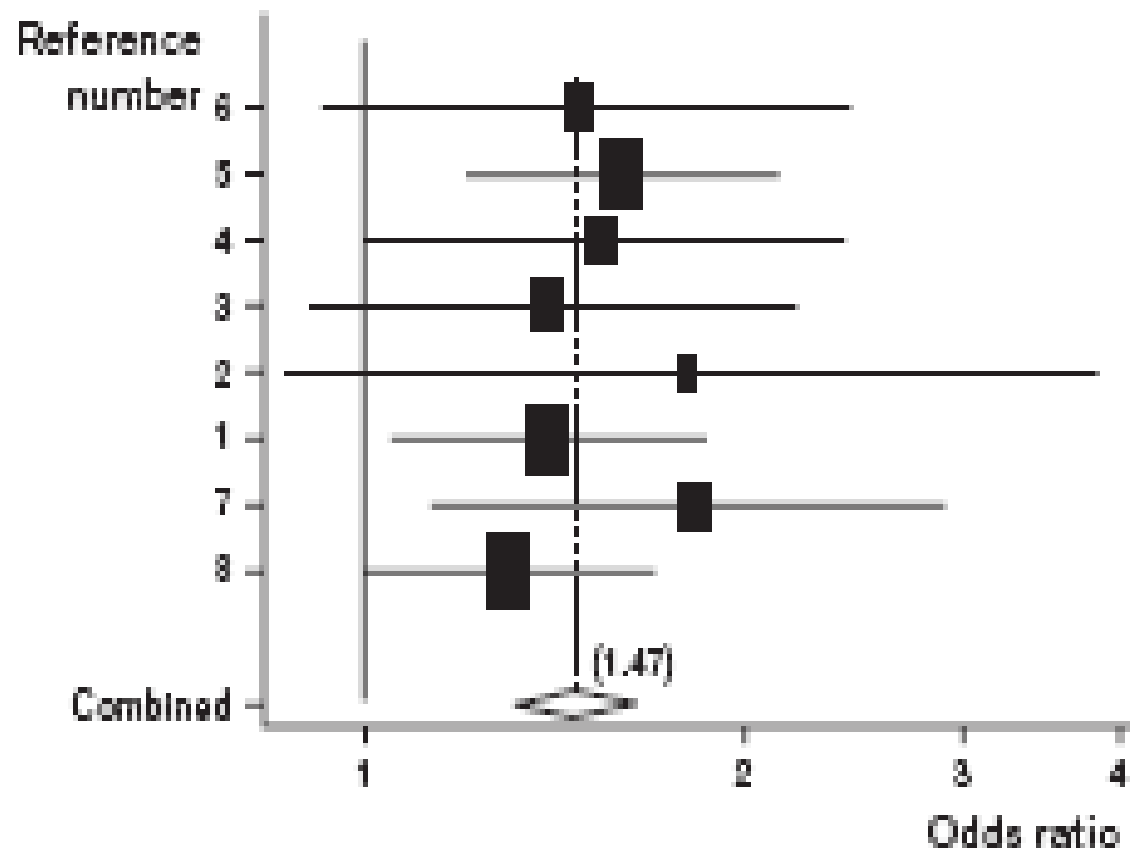
Epigenetic modification – changes in expression of DNA without alteration of DNA

Mechanisms:

- DNA methylation – decreases expression
- Micro RNA's – small sequence changes expression of RNA

Example : DRB1*1501 and adolescent obesity (OR 16)

Smoking is a risk factor for multiple sclerosis



Multiple Sclerosis: 3 Components

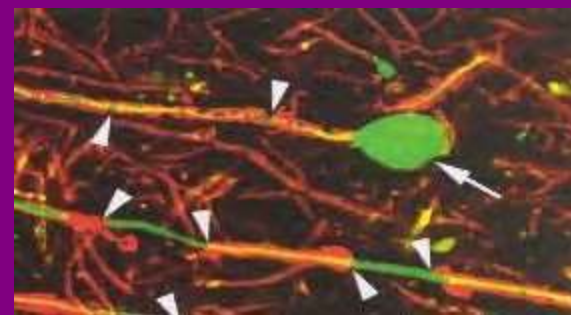
Inflammation



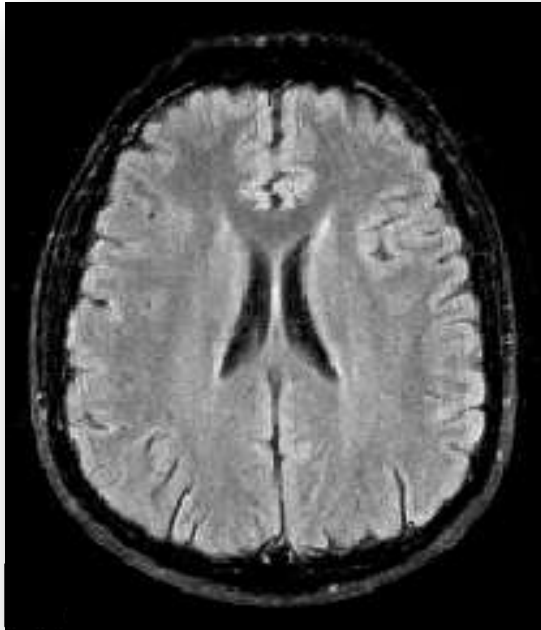
Demyelination



Axonal Loss



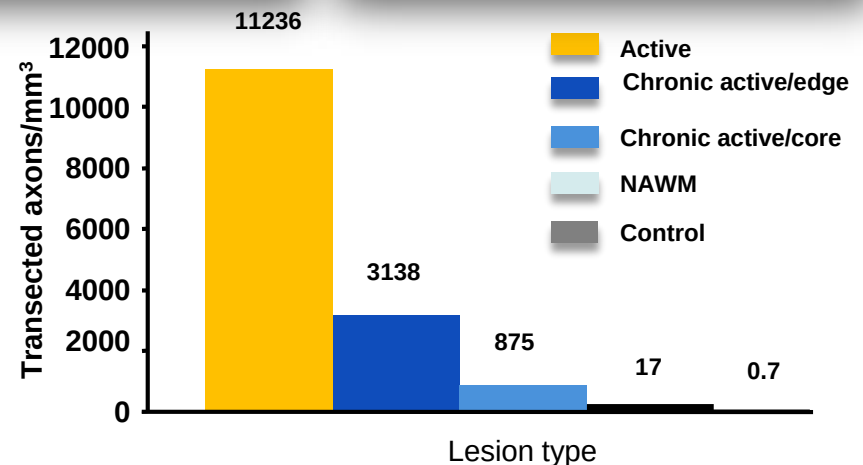
Inflammation causes Axonal Damage and Atrophy



Average reduction in brain volume¹

MS: 0.6 – 1.0% per year

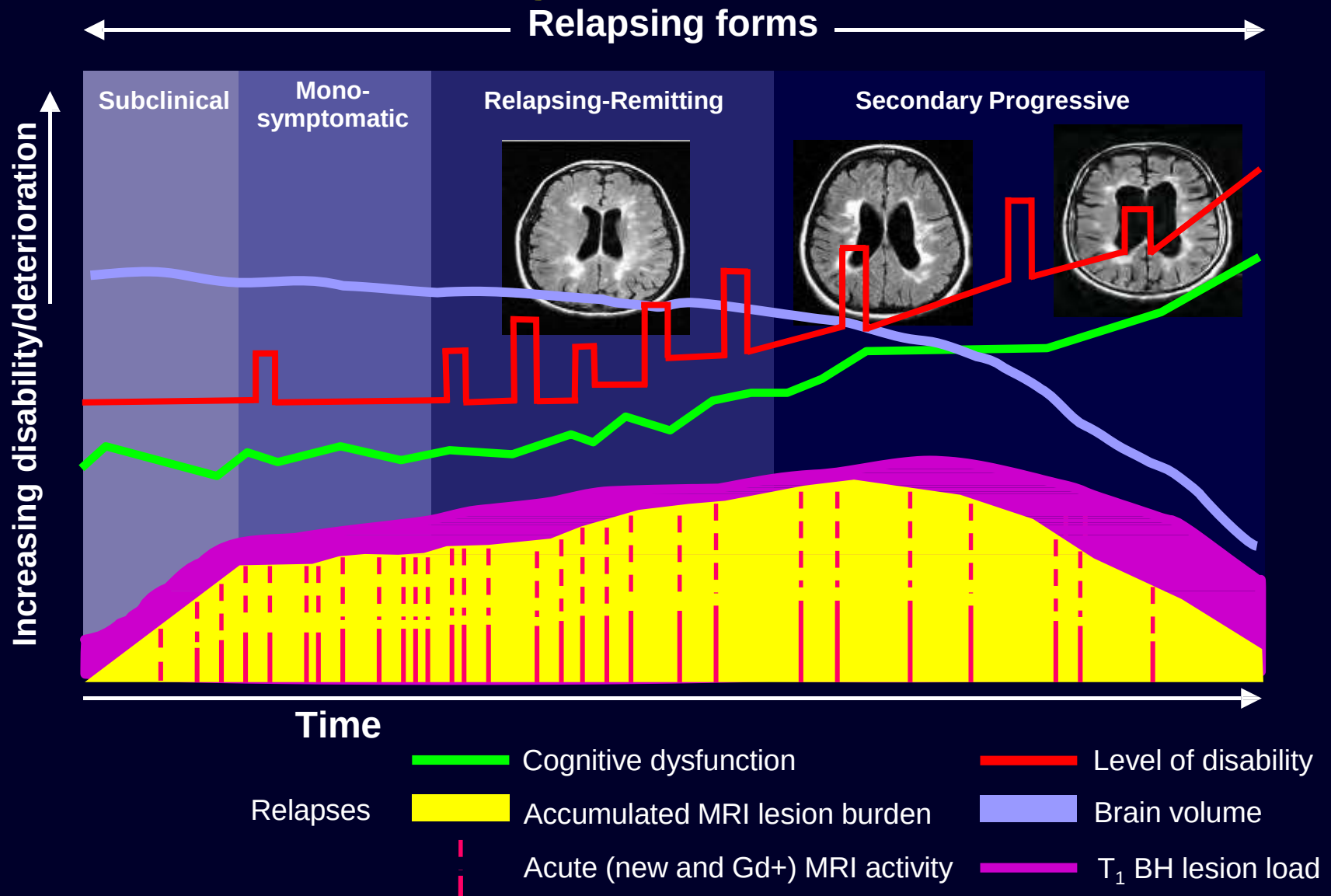
Healthy: 0.1 – 0.3% per year



NAWM=normal-appearing white matter

1. Miller DH et al. *Brain* 2002;125:1676-1695;
2. Trapp BD et al. *New Engl J Med.* 1998;338:278-285.

Natural History of MS

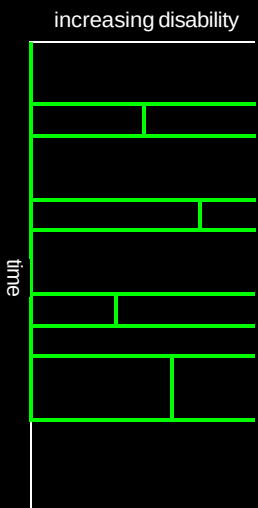


BH = black hole; Gd = gadolinium; MRI = magnetic resonance imaging.

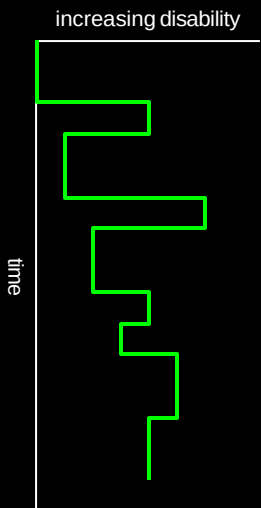
Noseworthy et al. *N Engl J Med.* 2000;343:938; Weinshenker et al. *Brain.* 1989;112:133; Trapp et al. *Curr Opin Neurol.* 1999;12:295.

Clinical courses of MS

(A)

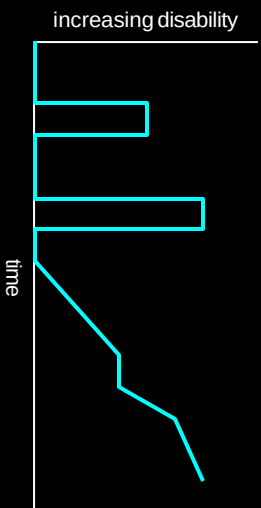


(B)

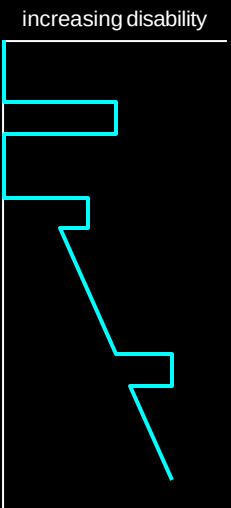


RR

(C)

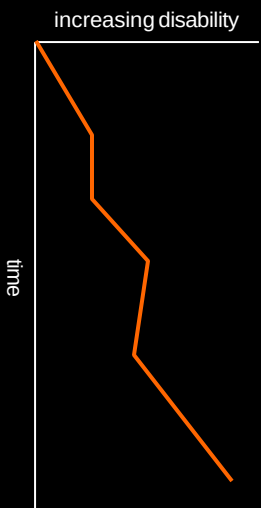


(D)

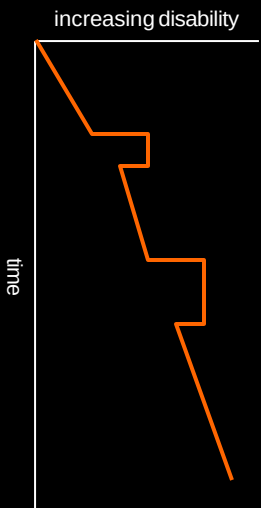


SP

(E)

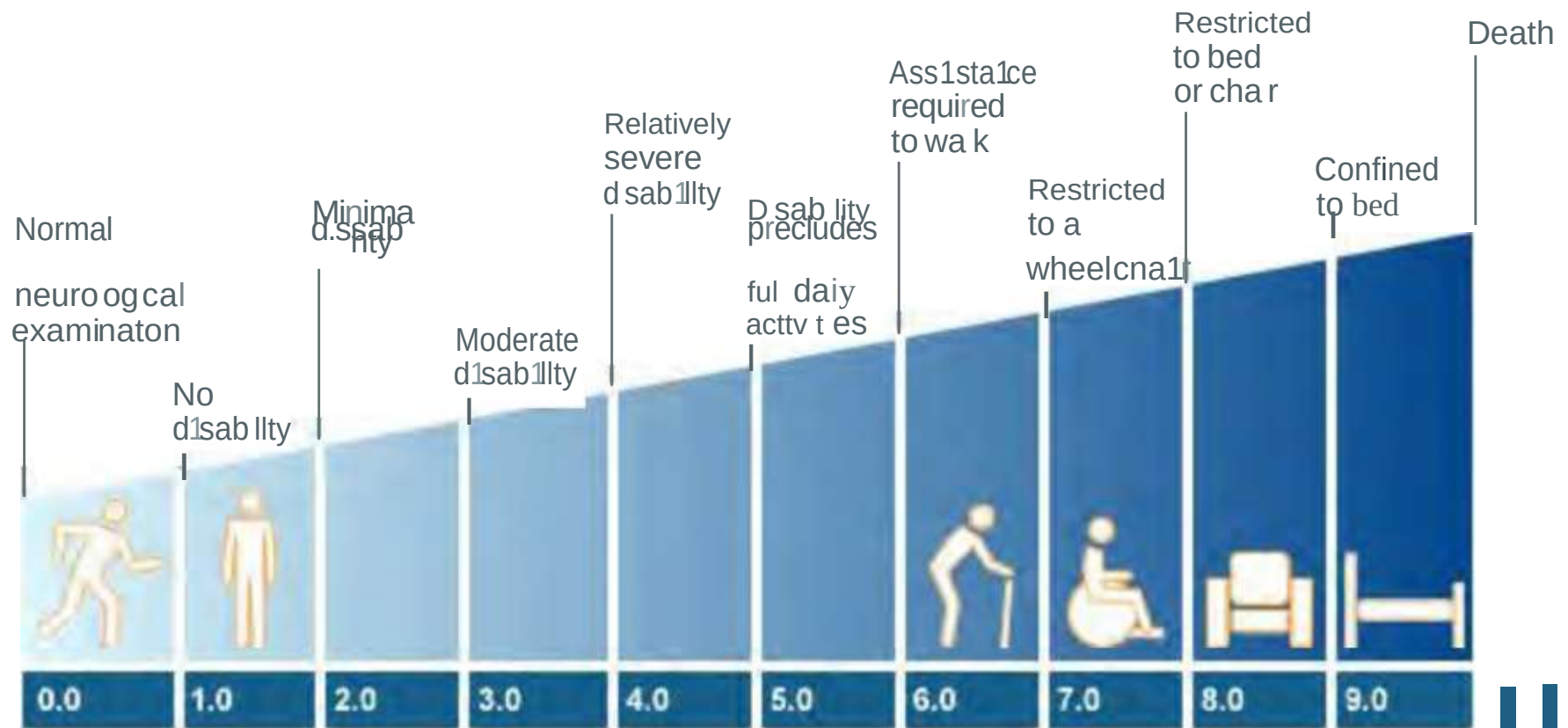


(F)



PP

MS Expanded Disability Status Scale - EDSS

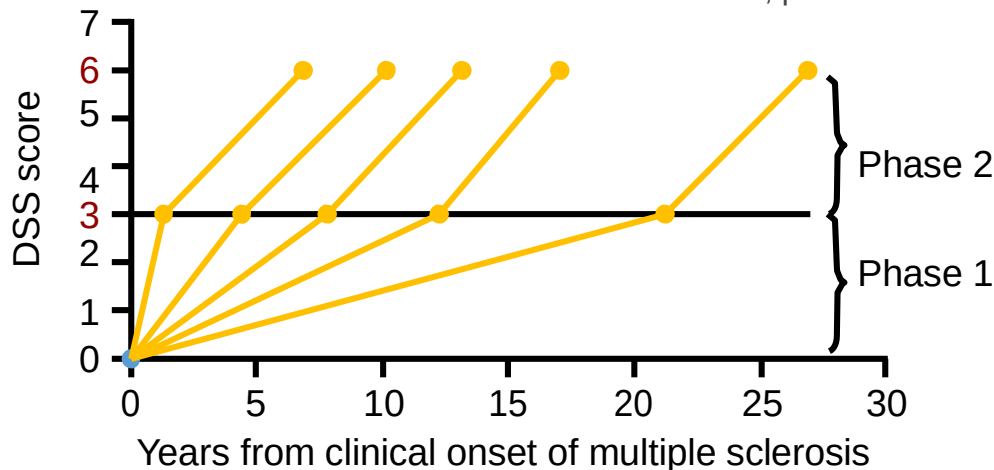


EDSS score <3.0 as a treatment goal: suggested approach illustrates limitations of EDSS

Traditionally, EDSS 4.0 has been seen as a milestone for disability progression, but it has been suggested that a new treatment goal may be to keep patients at an EDSS score of 3.0 or less¹

This arises from the school of thought that disability progression is a 2-stage process:¹

Phase 1 = time from onset to EDSS score 2.5; phase 2 = EDSS score 3.0–6.0



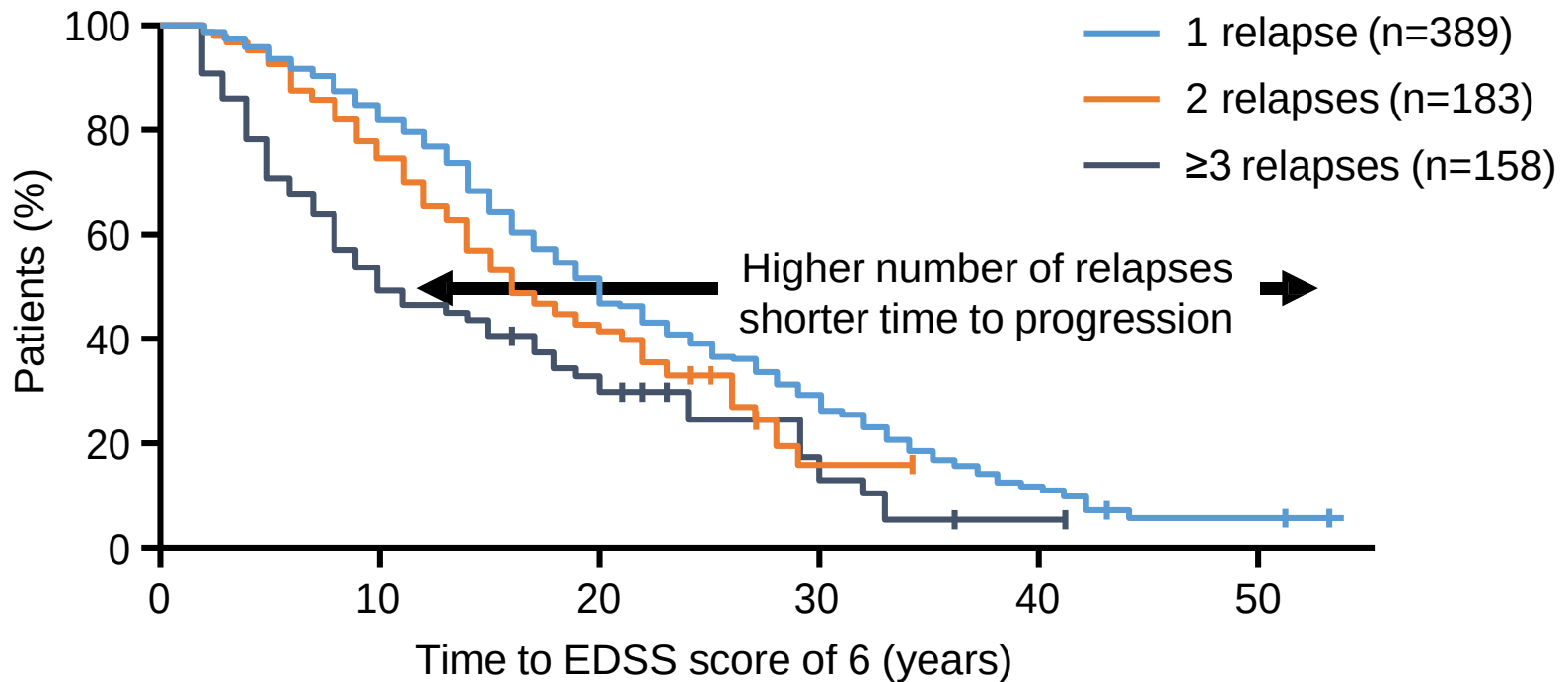
Disability progression in Phase 2 in five groups of patients (N=718) defined by duration of Phase 1¹

- However, the 2-stage distinction may be artifactual, caused by the nature of the EDSS itself
 - Over-reliance on motor function assessment prior to EDSS 7.0²

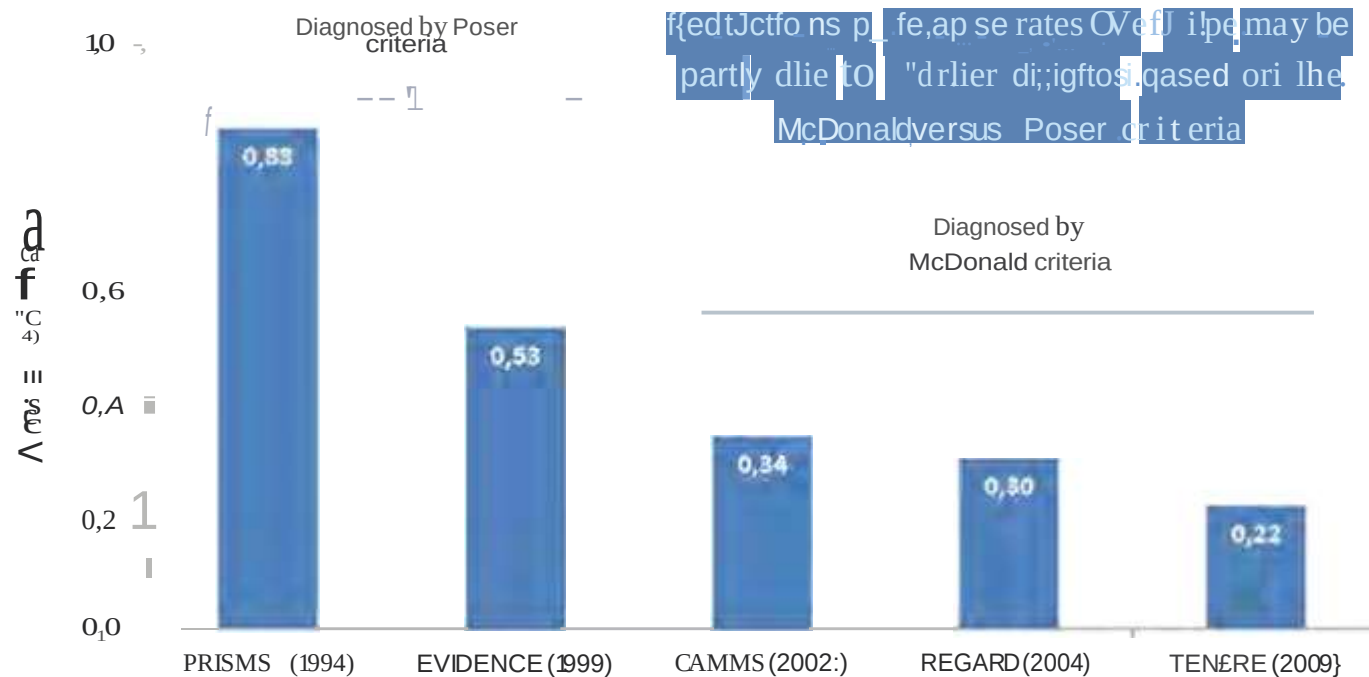
Early relapse activity is associated with long-term disability progression in natural history cohorts¹⁻⁵

Natural history data over a mean follow up of nearly 3 decades show that number of relapses in the first 2 years after diagnosis correlates with time to disability progression³

Stratification based on the number of relapses in first 2 years⁴



Rate of relapse has reduced over time with high dose of IFN beta in RRMS studies



Annualized relapse rates in the treatment arms of Rebif 44 µg subcutaneously three times weekly, across RRMS studies (with year that recruitment started)

Ulf Ciehaag BMJ et al. Curr Med Res Opin 2011; 27: 529-37

Prognosis for MS

‘after 15 years 50 % of MS patients will require an aid to walk (EDSS 6.0)’

Weinshenker et al. *Brain*. 1989;112:133, Runmarker and Anderson. *Brain*. 1993;116:117

Median EDSS after 14.1 years 3.25

Brex et al. *N Engl J Med*. 2002;346:158.

**Median time to EDSS 6 in European population of 2,495 patients
27.3 years**

Debouverie et al *Neurology* 2007;68:29

Prognosis for MS

Clinical predictors of disease course

Multifocal presentations

Interval between 1st and 2nd relapse

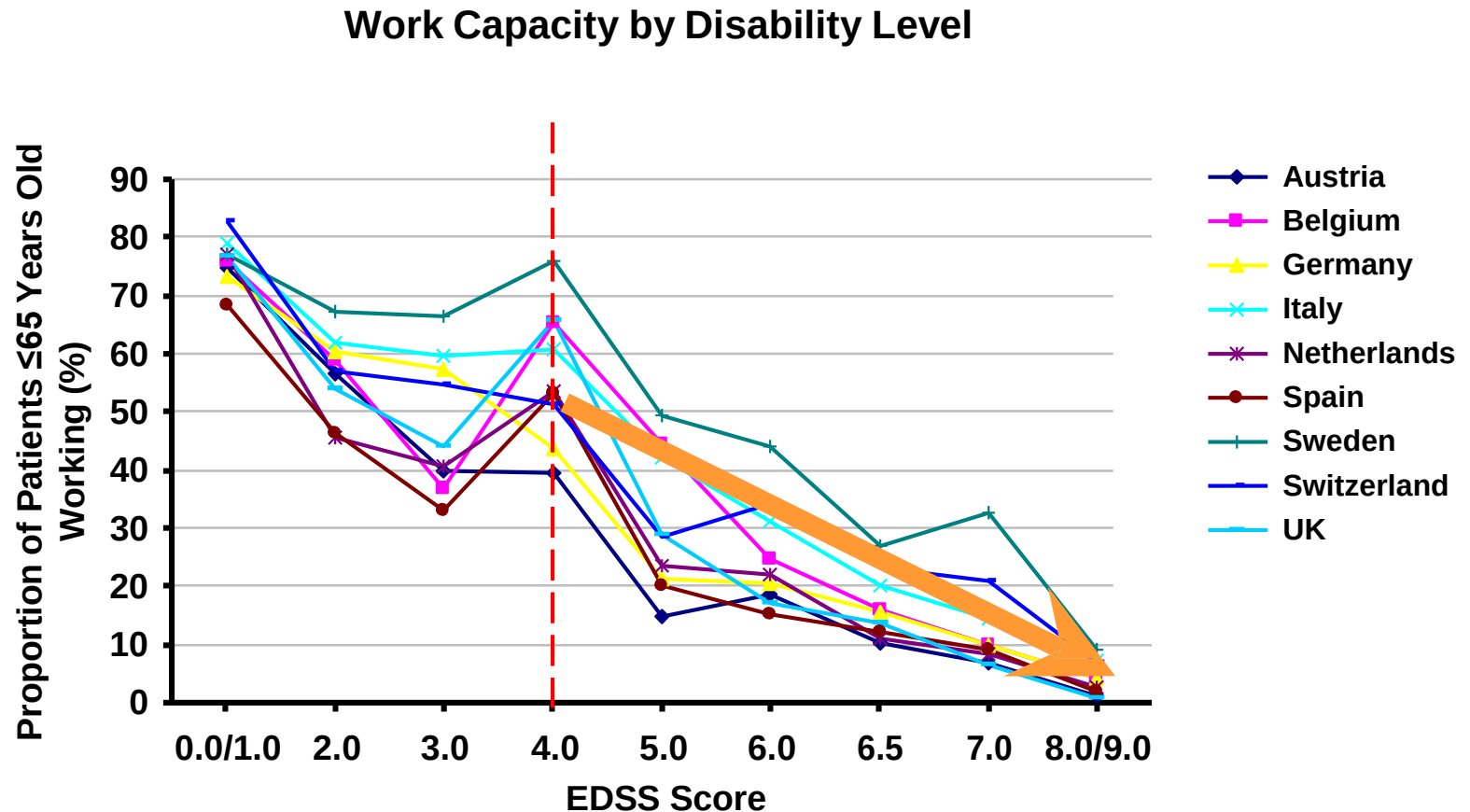
Relapse frequency

Relapse recovery

Progression of disability in the first five years of disease

Weinshenker et al. *Brain*. 1989;112:133, Runmarker and Anderson. *Brain*. 1993;116:117

Employment Change With EDSS Score



The proportion of patients employed or on long-term sick leave is calculated as a percentage of patients aged 65 or younger. The reduced work capacity particularly at EDSS 2.0–3.0 compared with EDSS 4.0 is explained by an increased frequency of relapses in this group and hence a concentration of sick leaves.

Discuss

Multiple sclerosis – with persisting symptoms

A definite diagnosis of Multiple Sclerosis by a Consultant Neurologist. There must be current clinical impairment of motor or sensory function, which must have persisted for a continuous period of at least 6 months.

Poser Diagnostic Criteria (1983)

Clinical Definite MS – clinically definite MS by Poser Criteria

2 attacks and one paraclinical evidence (MRI, evoked potentials)

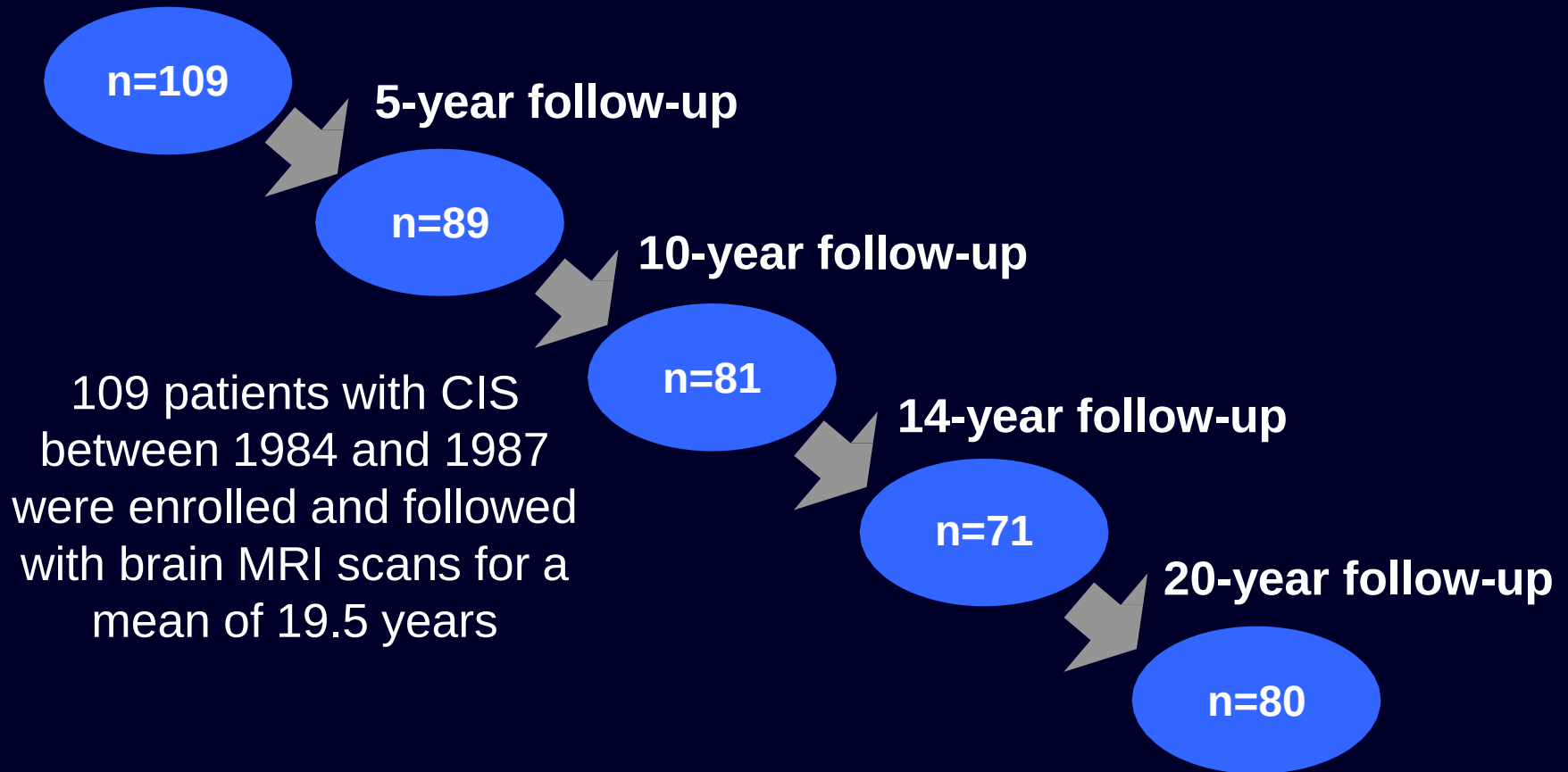
Laboratory-supported definite MS

1 attack and one paraclinical evidence and oligoclonal bands

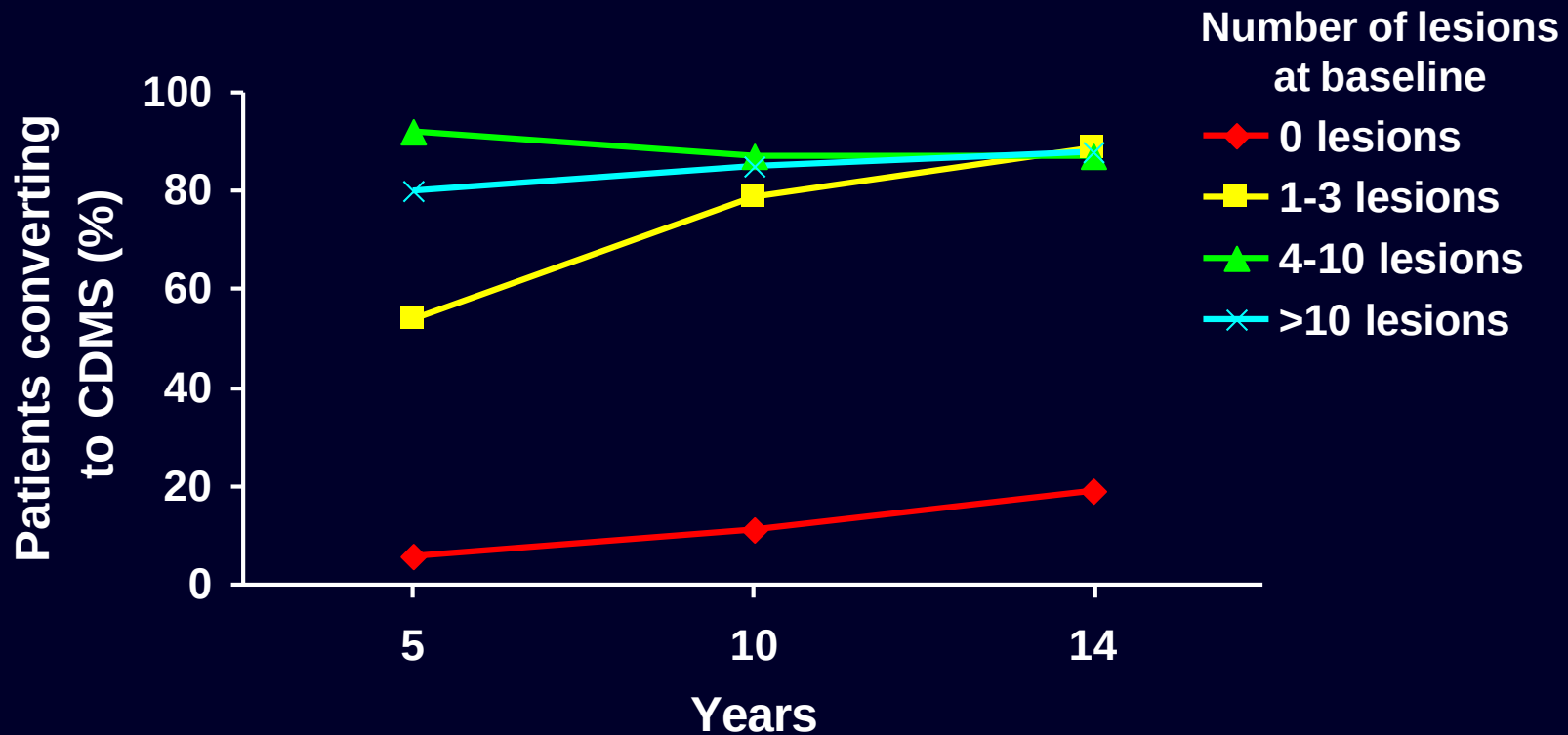
Clinically probable MS

Laboratory supported probable MS

Patients Presenting With CIS: The Queens Square Studies



Prognostic Significance of MRI in CIS: Rate of Conversion to CDMS



The data presented for years 5, 10, and 14 were obtained from different publications based on the same longitudinal study.

Morrissey et al. *Brain*. 1993;116:135; O'Riordan et al. *Brain*. 1998;121:495; Brex et al. *N Engl J Med*. 2002;346:158.

20-Year Follow-up in Patients With CIS

	baseline	5.3 yrs	14.1yrs	20yrs
No. of patients	109	89	71	87
CDMS (%)	0	38 (43)	48(68)	55(63)
CDMS abnormal MRI at CIS (%)	0	37 (65)	44 (88)	49 (86)
CDMS normal MRI at CIS (%)	0	1 (3)	4 (19)	6 (20)
EDSS CDMS cases (median)		2.5	3.25	40% < 3 40 % 7.5

Clinically Isolated Syndrome

Definition - a first neurological event suggestive of Multiple Sclerosis lasting for at least 24 hours and with symptoms and signs indicating a single or multiple lesions

e.g. – optic neuritis, spinal cord syndrome, brainstem lesion – INO

Sensory Symptoms –	34 %
Weakness	22 %
Ataxia	11 %
Visual symptoms	13 %
Diplopia	8 %
Bladder symptoms	1 %

Diagnostic Criteria

Diagnostic Criteria for Multiple Sclerosis: 2010 Revisions to the McDonald Criteria

Chris H. Peltman, MD, PhD,¹ Stephen C. Reingold, PhD,² Brenda Banwell, MD,³
Michel Canet, MD,⁴ Jeffrey A. Cohen, MD,⁵ Massimo Filippi, MD,⁶ Kazuo Fujihara, MD,⁷
Eva Havrdova, MD, PhD,⁸ Michael Hutchinson, MD,⁹ Ludwig Kappes, MD,¹⁰
Fred D. Lublin, MD,¹¹ Xavier Montalban, MD,¹² Paul O'Connor, MD,¹³
Magnus Sandberg-Vollheim, MD, PhD,¹⁴ Alan J. Thompson, MD,¹⁵
Emmanuelle Waubant, MD, PhD,¹⁶ Brian Weinshenker, MD,¹⁷ and Jerry S. Wolinsky, MD¹⁸

New evidence and consensus has led to further revision of the McDonald Criteria for diagnosis of multiple sclerosis. The use of imaging for demonstration of dissemination of central nervous system lesions in space and time has been simplified, and in some circumstances dissemination in space and time can be established by a single scan. These revisions simplify the Criteria, preserve their diagnostic sensitivity and specificity, address their applicability across populations, and may allow earlier diagnosis and more uniform and widespread use.

ANN NEUROL 2011;69:292-302

DIS

2010 criteria



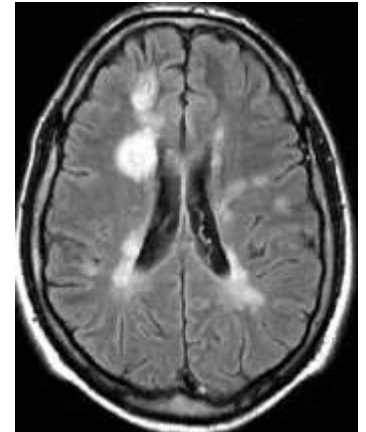
spinal



juxta



infra



PV

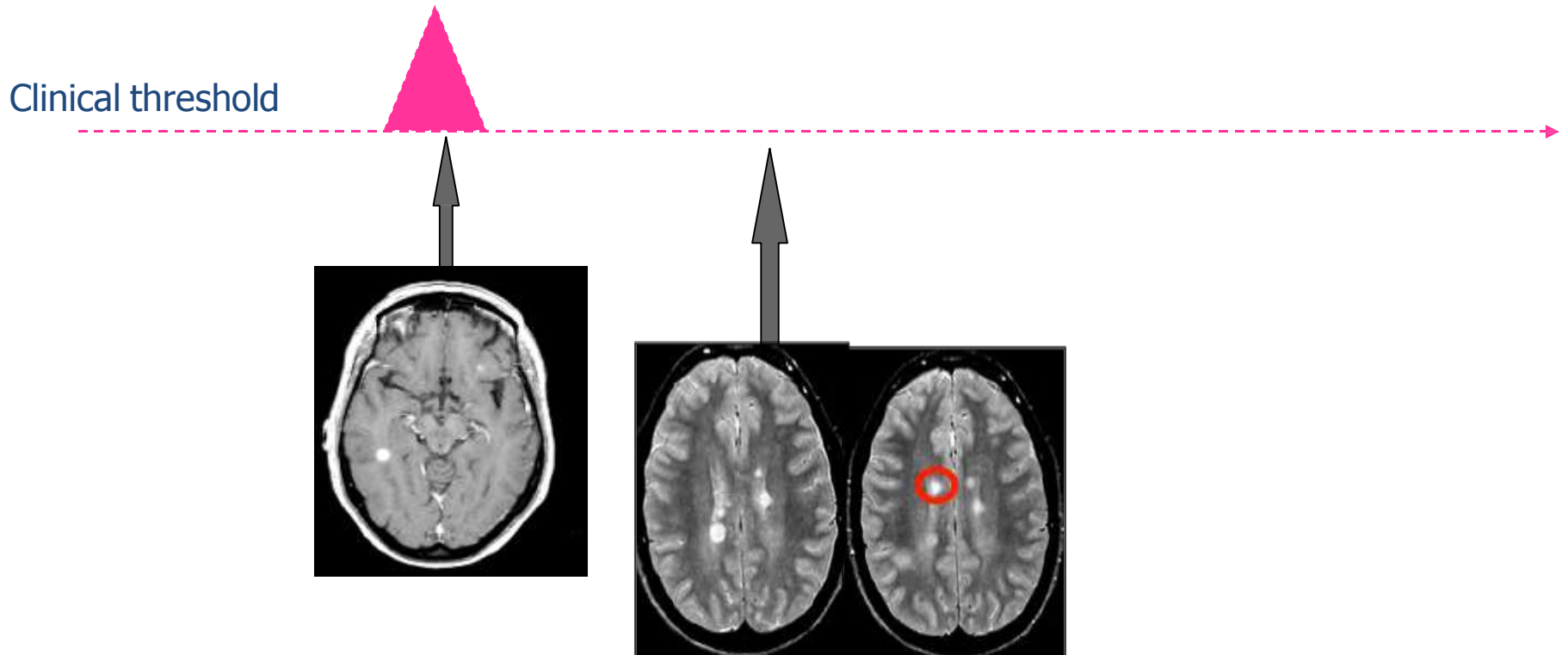
≥ 1 asymptomatic T2 lesion in ≥ 2 topographies

DIT

2010 criteria

A new T2 on follow-up MRI, with reference to a baseline scan, irrespective of the timing of the baseline MRI

Simultaneous presence of asymptomatic gadolinium-enhancing and nonenhancing lesions at any time



New McDonald Criteria 2010

Dissemination in Space (DIS)

≥ 1 T2 lesion in at least 2 out of four MS-typical regions of CNS

periventricular,

juxtacortical,

infratentorial,

or spinal cord

does not include 9 T2 lesion cut-off

Dissemination in time (DIT)

Simultaneous presence of asymptomatic Gd-enhancing AND non-enhancing lesions at any time

Or:

A new T2 and/or Gd+ lesion(s) on FU MRI irrespective of timing of baseline scan

MRI focal white matter lesions

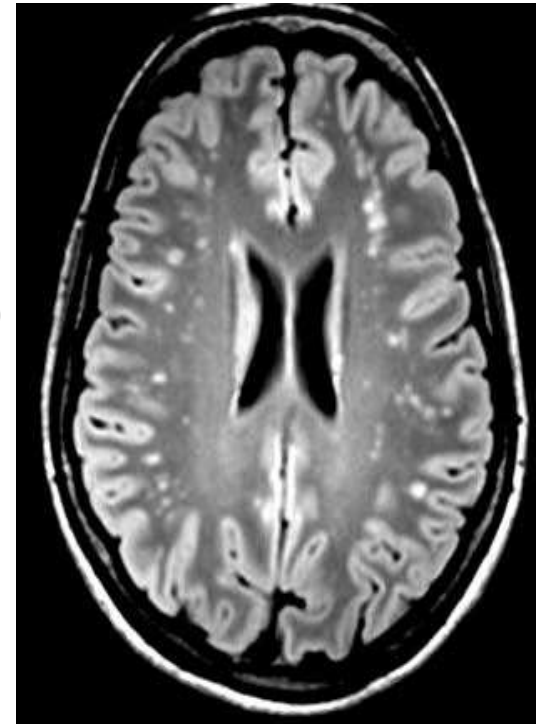
Incidental focal white matter lesions are far more common than **multiple sclerosis**

Multiple sclerosis:

Prevalence **0.1/0.2%** (20-40 years)

MRI focal white matter lesions (incidental):

Prevalence **5-10%** (20-40 years)

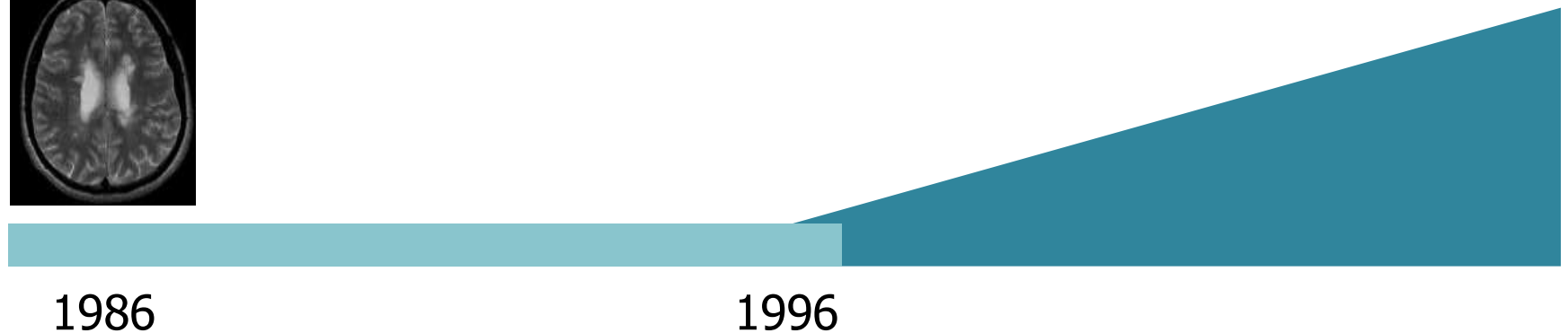
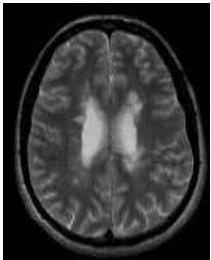


Preclinical PPMS

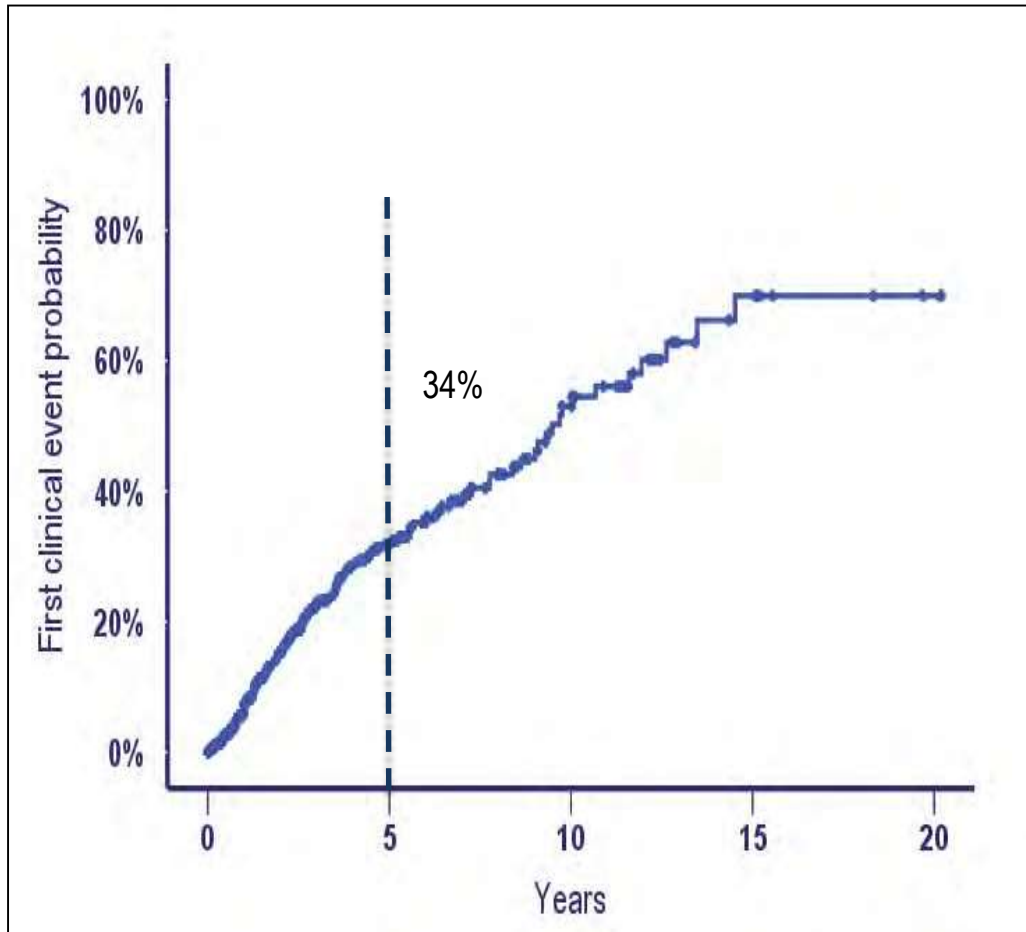
Clinical presentation of primary progressive multiple sclerosis 10 years after the incidental finding of typical magnetic resonance imaging brain lesions

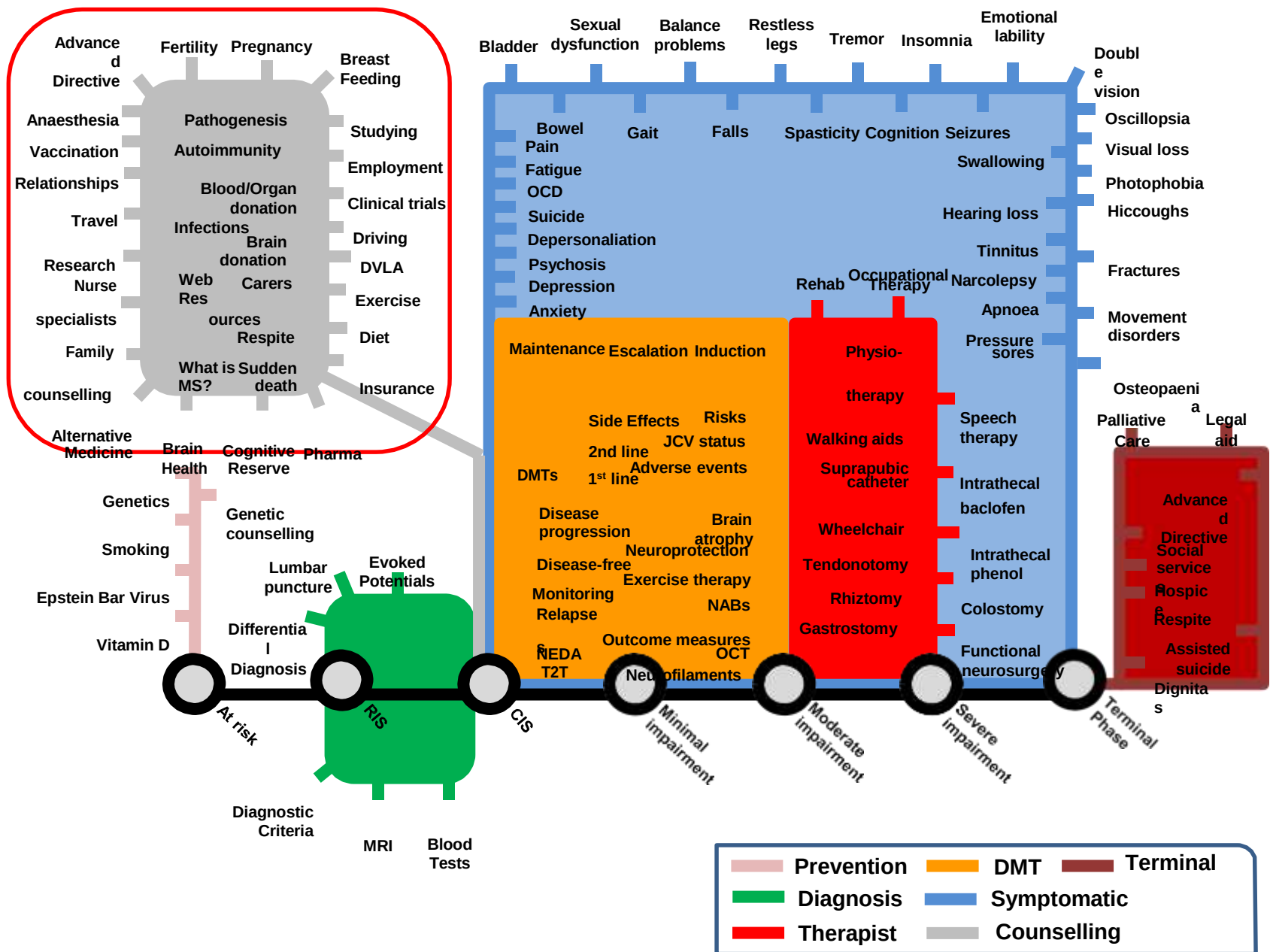
The subclinical stage of primary progressive multiple sclerosis may last 10 years

39y healthy control for a
Parkinson's disease study

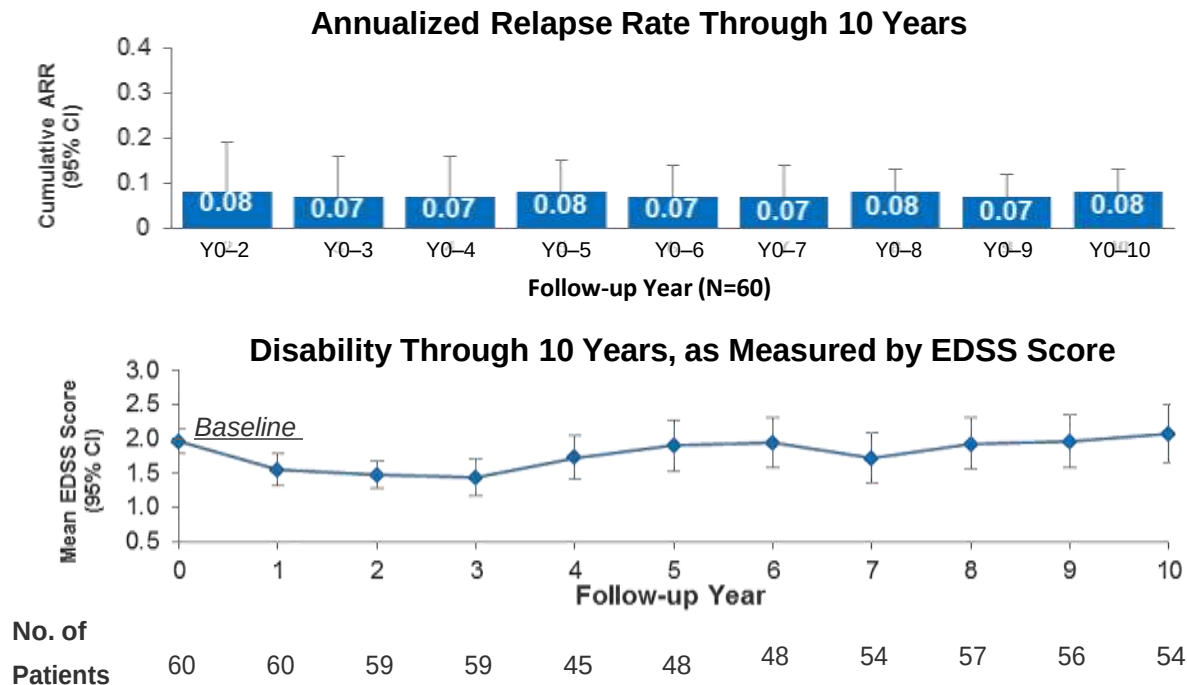


Evolution to CDMS at 5-years





[P679] Efficacy and Safety of Alemtuzumab in Patients With RRMS is Durable Over 10 Years: Follow-up From the CAMMS223 Study (Selected Data)



- At CAMMS223 baseline, 100% of patients had Gd-enhancing lesions
 - After alemtuzumab treatment, the proportion of patients with Gd-enhancing lesions was low (CARE-MS extension baseline: 8.3%; CARE-MS extension Years 1, 2, and 3 were 5.6%, 13.0%, and 5.5%, respectively)

Selmaj K et al. ECTRIMS 2016, Poster P679.



Thank You

